Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Содержание (i)

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
bgaryaev@bgaryaev:~$ ls /etc/ > file.txt
bgaryaev@bgaryaev:~$ ls >> file.txt
bgaryaev@bgaryaev:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
```

Рисунок 1: Запись в файл

Работа фильтра

```
bgaryaev@bgaryaev:~$ grep .conf file.txt > conf.txt
bgaryaev@bgaryaev:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
```

Рисунок 2: Поиск расширения .conf

Команда поиска

```
/home/bgaryaev/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage03/
report/bib/cite.bib
/home/bgaryaev/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage04/
report/_resources/csl
/home/bgaryaev/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage04/
report/bib/cite.bib
/home/bgaryaev/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage05/
report/_resources/csl
/home/bgaryaev/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage05/
report/bib/cite.bib
/home/bgaryaev/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage06/
report/_resources/csl
/home/bgaryaev/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage06/
report/bib/cite.bib
/home/bgaryaev/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/_reso
urces/csl
/home/bgaryaev/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/bib/c
ite.bib
/home/bgaryaev/git-extended/.git/hooks/commit-msg.sample
/home/bgaryaev/git-extended/.git/objects/05/c10434ef58a4c7b779cee408cddae32298bce0
/home/bgaryaev/git-extended/.git/objects/87/ca2325f658cfad49e2c00ae3d5a13b9b5c0a2b
/home/bgaryaev/git-extended/.git/objects/87/c6ad8bdc488c2b269759ccbe7d7ac45ddddaebd
/home/bgaryaev/git-extended/.git/objects/af/c97d90828c055f37e5d3a94886127777fd4fe5
/home/bgaryaev/git-extended/.git/config
/home/bgaryaev/conf.txt
bgaryaev@bgaryaev:~$
```

Рисунок 3: Поиск файлов

Команда поиска

```
/etc/foomatic/hashes.d/hashes.new
find: /etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
'/etc/libvirt': Permission denied
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
:find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sssd': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

Управление процессами

```
bgaryaev@bgaryaev:~$
bgaryaev@bgaryaev:~$ find ~ -name "log*" > logfile &
[1] 6299
bgaryaev@bgaryaev:~$
[1]+  Done                    find ~ -name "log*" > logfile
bgaryaev@bgaryaev:~$ rm logfile
bgaryaev@bgaryaev:~$
```

Рисунок 5: Фоновый запуск процесса

Управление процессами

```
bgaryaev@bgaryaev:~$
bgaryaev@bgaryaev:~$ gedit &
[1] 6335
bgaryaev@bgaryaev:~$ ps | grep gedit
   6335 pts/1    00:00:00 gedit
bgaryaev@bgaryaev:~$ kill 6335
[1]+  Terminated              gedit
bgaryaev@bgaryaev:~$
bgaryaev@bgaryaev:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

Команды df и du

```
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This  manual  page  documents the GNU version of df.  df displays the amount of space available on
       the file system containing each file name argument.  If no file name is given, the space available
       on all currently mounted file systems is shown.  Space is shown in 1K blocks  by  default,  unless
       the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

       If  an  argument  is  the absolute file name of a device node containing a mounted file system, df
       shows the space available on that file system rather than on the file system containing the device
       node.  This version of df cannot show the space available on unmounted file  systems,  because  on
       most kinds of systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
       Show information about the file system on which each FILE resides, or all file systems by default.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

       -B, --block-size=SIZE
              scale  sizes  by  SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576
              bytes; see SIZE format below

       --direct
 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

Команды df и du

```
DU(1)                                User Commands                                DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print  apparent  sizes  rather  than  device  usage;  although the apparent size is usually
              smaller, it may be larger due to holes in ('sparse') files, internal  fragmentation,  indi-
              rect blocks, and the like

       -B, --block-size=SIZE
              scale  sizes  by  SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576
              bytes; see SIZE format below

       -b, --bytes
              equivalent to '--apparent-size --block-size=1'

 Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

Команды df и du

```
bgaryaev@bgaryaev:~$
bgaryaev@bgaryaev:~$ df
Filesystem     1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 89449364  65063868  58% /
devtmpfs         4011280       0   4011280   0% /dev
tmpfs            4035652      92   4035560   1% /dev/shm
tmpfs            1614264    1952   1612312   1% /run
tmpfs               1024       0      1024   0% /run/credentials/systemd-journald.service
tmpfs            4035656    1592   4034064   1% /tmp
/dev/nvme0n1p3 155186176 89449364  65063868  58% /home
/dev/nvme0n1p2   1992552  518800   1352512  28% /boot
/dev/loop0         68480   68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop2         49280   49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop1        115712  115712         0 100% /var/lib/snapd/snap/hugo/25676
tmpfs               1024       0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             807128     172    806956   1% /run/user/1023
/dev/sr0         2677920 2677920         0 100% /run/media/bgaryaev/Fedora-WS-Live-43
bgaryaev@bgaryaev:~$
```

Рисунок 9: Справка по команде du

Команды df и du

```
4         ./git-extended/.git/objects/30
4         ./git-extended/.git/objects/10
4         ./git-extended/.git/objects/e3
4         ./git-extended/.git/objects/52
4         ./git-extended/.git/objects/e9
4         ./git-extended/.git/objects/d0
76        ./git-extended/.git/objects
8         ./git-extended/.git/logs/refs/heads
8         ./git-extended/.git/logs/refs/remotes/origin
8         ./git-extended/.git/logs/refs/remotes
16        ./git-extended/.git/logs/refs
20        ./git-extended/.git/logs
212       ./git-extended/.git
220       ./git-extended
0         ./monthly
0         ./reports/monthly/monthly
0         ./reports/monthly
0         ./reports
4         ./ski.plases/equipment
0         ./ski.plases/plans
4         ./ski.plases
0         ./australia
0         ./play/games/play
0         ./play/games
0         ./play
842600    .
bgaryaev@bgaryaev:~$
```

Рисунок 10: Запуск команды du

Команда поиска

```
/home/bgaryaev/git-extended/.git/objects/87
/home/bgaryaev/git-extended/.git/objects/20
/home/bgaryaev/git-extended/.git/objects/af
/home/bgaryaev/git-extended/.git/objects/30
/home/bgaryaev/git-extended/.git/objects/10
/home/bgaryaev/git-extended/.git/objects/e3
/home/bgaryaev/git-extended/.git/objects/52
/home/bgaryaev/git-extended/.git/objects/e9
/home/bgaryaev/git-extended/.git/objects/d0
/home/bgaryaev/git-extended/.git/logs
/home/bgaryaev/git-extended/.git/logs/refs
/home/bgaryaev/git-extended/.git/logs/refs/heads
/home/bgaryaev/git-extended/.git/logs/refs/remotes
/home/bgaryaev/git-extended/.git/logs/refs/remotes/origin
/home/bgaryaev/monthly
/home/bgaryaev/reports
/home/bgaryaev/reports/monthly
/home/bgaryaev/reports/monthly/monthly
/home/bgaryaev/ski.plases
/home/bgaryaev/ski.plases/equipment
/home/bgaryaev/ski.plases/plans
/home/bgaryaev/australia
/home/bgaryaev/play
/home/bgaryaev/play/games
/home/bgaryaev/play/games/play
bgaryaev@bgaryaev:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.